FIG. 4 Verifier-Anchored Contextual Bandit Routing

(a) Deterministic UCB1 Policy

1. Direct untried? DIRECT
Explore direct implementer first

2. Parallel untried? PARALLEL
Explore 3-worker coding next

3. Both arms tried? ARGMAX UCB1
Deterministic score; ties → Direct

UPPER CONFIDENCE BOUND (src/policy.rs)

$$score(a) = s_a/n_a + \sqrt{(2 \ln N / n_a)}$$

Per context · max 1,024 contexts · atomic save

ZEUS JUDGE (SHADOW-ONLY OBSERVATION)
Logs pre-run features & typed Rank/Abstain in shadow-decision.json; never alters UCB1.

(b) Verifier Reward Gate (14 Field Runs)

Verified Pass	+1 trial, +1 success
Verified Fail	+1 trial, +0 success
Infra / Timeout / Cancel	ZERO policy update

14 REAL TRIALS (research/data/field_runs.json)



Outer ring = reward recorded (4/4 verified)
Error isolation: 0/10 errors touched UCB
○ Direct implementer □ Parallel workers

Guarantees network/LLM/sandbox failures never bias strategy selection toward the surviving arm.

(c) Persisted Router States

ACCEPTANCE NEXT: DIRECT
ctx: python-cli-reconstruction
rule: tie 2.18

DIR	1/1 (100%)
PAR	1/1 (100%)

MODEL-TEST NEXT: PARALLEL
ctx: python-cli-reconstruction
rule: parallel untried

DIR	1/1 (100%)
PAR	0/0 (0%)

ROUTER-TRAINING NEXT: PARALLEL
ctx: qwen3.5-9b-python-cli-persistence-v1
rule: parallel untried

DIR	1/1 (100%)
PAR	0/0 (0%)

(b)–(c) extracted verbatim from research/data/field_runs.json. Next choices recomputed with src/policy.rs rules.